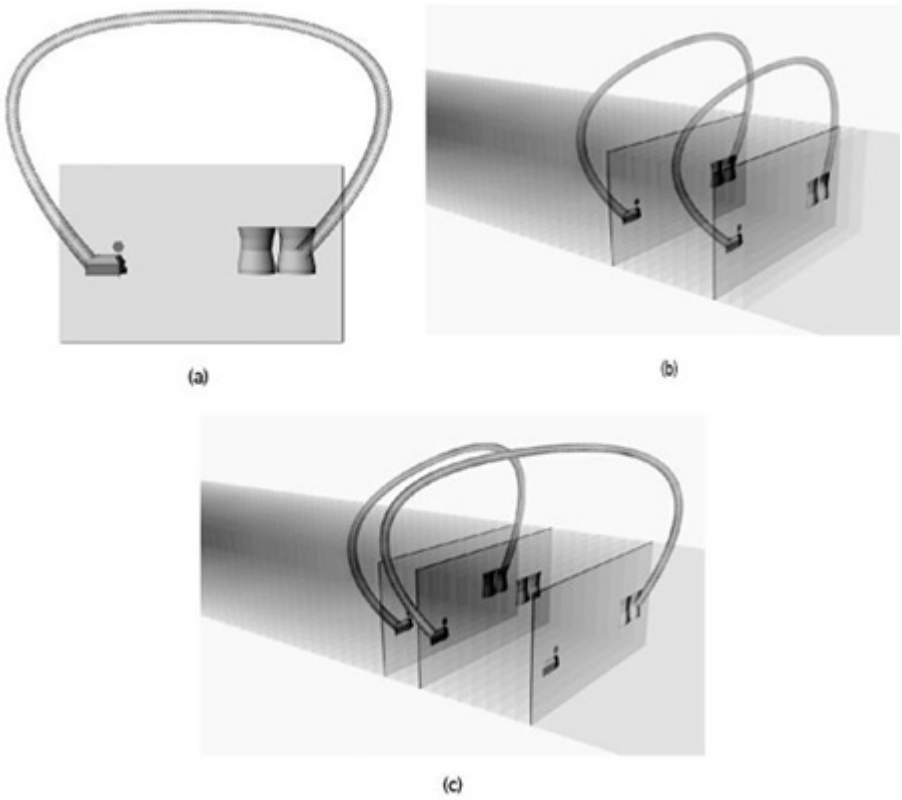


Figure 15.5 (a) A wormhole, created at some moment in time, connects one location in space with another. (b) If the wormhole mouths do not move relative to one another, they “pass” through time at the same rate, so the tunnel connects the two regions at the same time. (c) If one wormhole mouth is taken on a round-trip journey (not shown), less time will elapse for that mouth, and hence the tunnel will connect the two regions of space at different moments of time. The wormhole has become a time machine.

How about in practice? Well, as I mentioned at the outset, no one knows whether wormholes even exist. Some physicists have suggested that tiny wormholes might be plentiful in the microscopic makeup of the spatial fabric, being continually produced by quantum fluctuations of the gravitational field. If so, the challenge would be to enlarge one to macroscopic size. Proposals have been made for how this might be done, but they’re barely beyond theoretical flights of fancy. Other physicists have envisioned the creation of large wormholes as an engineering project in applied general relativity. We know that space responds to the distribution of matter and energy, so with sufficient control over matter and energy, we might cause a region of space to spawn a wormhole. This approach presents an additional complication, because just as we must